Nuclear Fission Plant

Sustainable Nuclear Energy

Assignment 1

Candidate:
Bartolomeo Arrabito, 349377
Fabio Geraci, 348108
Matteo Sabina, 337628

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Summary

1 General introduction

The aim of this study is to become familiar with fundamental concepts and key calculation methodologies related to natural circulation, a phenomenon particularly relevant in the nuclear field. Under standard operational conditions, coolant circulation within nuclear power plants is primarily driven by circulation pumps. However, buoyancy-driven natural circulation represents an essential passive safety mechanism, particularly in accidental scenarios when these pumps are unavailable or fail to operate. This study specifically addresses such an incidental scenario, in which coolant flow relies entirely on natural forces generated by density variations due to thermal gradients.

Therefore, understanding natural circulation phenomena becomes fundamental to ensuring the plant's safety and reliability during off-normal or emergency situations

2 Assignment 1.1

The analysis is carried out in a simplified system, as it can be seen in Figure 1, a rectangular loop where natural circulation is the only driver between the heater and the cooler. The heater transfers energy to the fluid, bringing it from saturated liquid to saturated vapor, while the refrigerator condenses the vapor to saturated liquid. The circuit is divided in two branches, one hot uphill where vapor goes through and one cold downhill where liquid flows. The elevation difference between the heater and the cooler generates the buoyancy force that drives the circulation.

The main goal of the exercise is to find the elevation difference that enables natural circulation to fully remove the specified thermal power. In a second step, the pipe length is treated as a free parameter with the additional constraint of having the total pipes length equal to the elevation difference, and finally, the influence of pipe roughness is evaluated by comparing three different surface conditions.

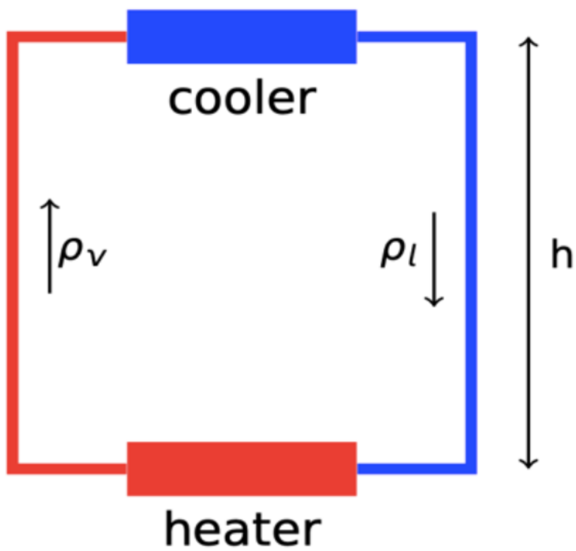

Figure 1: Scheme of the natural circulation loop

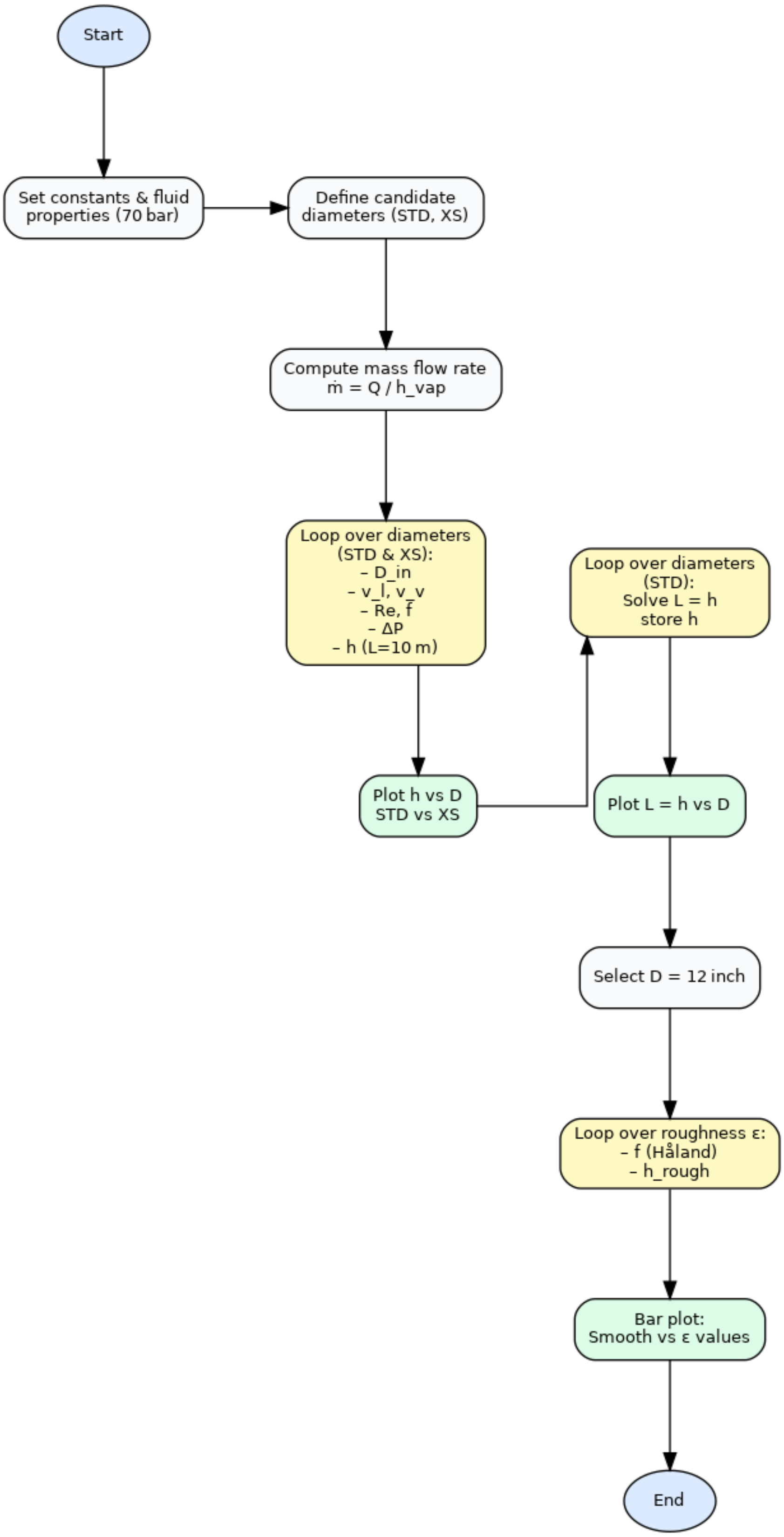

Figure 2: Flow chart

2.1 Geometry and Data

For the analysis of this rectangular loop, both the heater and cooler are placed horizontally, the system operates in steady-state, heat transfer along the vertical legs is neglected, and both liquid and vapor phases remain at saturation conditions at constant pressure (70 bar).

It has provide, as data, a thermal power of 34.8 MW_{th}, the length of every side of the pipe equal to 10 m, and specific localized pressure loss coefficients ($k = 20$ for the cooler and $k = 40$ for the heater). In addition, the pipe diameters are constrained by standard industrial values, and the pipe roughness is set to $\epsilon = 2.5 \times 10^{-5}$ m and $\epsilon = 5.0 \times 10^{-5}$ m, as provided in Figure 7.

2.2 Assumption

The theoretical framework is established by first applying the conservation equations for mass and momentum:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0 \tag{1}$$

Where ρ is the density of the fluid and v the velocity.

$$\frac{\partial (\rho v)}{\partial t} = -\nabla p + \nabla \cdot \tau + \rho \mathbf{g} \tag{2}$$

In which p represents the internal pressure, τ represent the shear stresses and g is the gravitational acceleration.

Under the following assumptions:

1. **Steady-state flow:** the time-derivatives vanish ($\partial/\partial t = 0$), and the material derivative becomes

$$\rho \frac{Dv}{Dt} = \nabla \cdot (\rho v v). \tag{3}$$

2. **Incompressible fluid:** the density is constant, so Eq. (1) reduces to

$$\nabla \cdot v = 0. \tag{4}$$

3. **One-dimensional momentum integral:** projecting along the loop coordinate l, the momentum equation simplifies to

$$\frac{\partial}{\partial l}\left(\frac{G^2}{\rho}\right) = -\frac{\partial P}{\partial l} - \frac{\partial P}{\partial l}\Big|_{\text{friction}} - \rho g \cos\theta, \tag{5}$$

where $G = \rho v$ is the mass velocity, and θ is the local inclination of the flow path.

The analysis begins with the steady-state momentum equation, integrated around a closed circulation loop:

$$\oint \frac{d}{dl}\left(\frac{G^2}{\rho}\right) dl = 0 \qquad \text{(no variation of cross-sectional area)} \tag{6}$$

$$\oint \frac{\partial P}{\partial l} dl = 0 \qquad \text{(loop is closed)} \tag{7}$$

The momentum equation along the loop becomes:

$$\oint \frac{dP}{dl}\Big|_{\text{friction}} dl = \frac{1}{2} f \frac{L}{D} \frac{G^2}{\rho} + \sum_i k_i \frac{AG^2}{2\rho} \tag{8}$$

Where the first term on the right-hand side represents the **distributed pressure drop** and the second term represents the **localized losses** (e.g., heater, cooler, bends).

The contribution from the gravity term along the vertical coordinate is:

$$\oint \rho g \cos(\theta) dl = \oint \rho g dz = \rho_H g(z_b - z_a) + \rho_C g(z_d - z_c) \tag{9}$$

which gives the **buoyancy driving force**:

$$(\rho_H - \rho_C) g H = \Delta P_{\text{buoyancy}} \tag{10}$$

In which H represent the net elevation.

Obtaining the net elevation:

$$H = \frac{\Delta P_{\text{buoyancy}}}{(\rho_H - \rho_C) g} \tag{11}$$

2.3 Analysis and Results

The analysis considered three distinct cases:

- Variable elevation, diameter optimization and fixed geometry comparison

  Pipe length was fixed; instead, the two commercially available pipe schedules Standard (STD) and Extra Strong (XS) were compared to determine the elevation head required to offset the frictional losses in each.

- Fixed pipe length , diameter variation

  Pipe diameters were varied to identify the optimal elevation head.

- Surface-roughness comparison A smooth-walled (ideal) pipe was compared against two commercially available roughness classes to assess the impact of wall roughness on head loss.

2.3.1 Pipe height evaluation

In order to assess the required elevation head for a pipe of maximum length 10 m to dissipate the prescribed thermal power by natural circulation, a MATLAB code was developed. This approach involves an iterative procedure over a set of a standardized pipe sizes, according to ANSI B36.10 (Figure 7). For each nominal outer diameter, two different wall thicknesses were considered Standard (STD) and Extra Strong (XS) to assess how the choice of pipe schedule influences hydraulic performance. This allowed us for have a better comparison between configurations with the same outer diameter but different internal ones. At each iteration, the velocities of the liquid and vapor phases were computed. The corresponding pressure drop was then evaluated, and the minimum elevation head, required to counterbalance it with the buoyancy forces arising from the density difference between the phases, was determined.

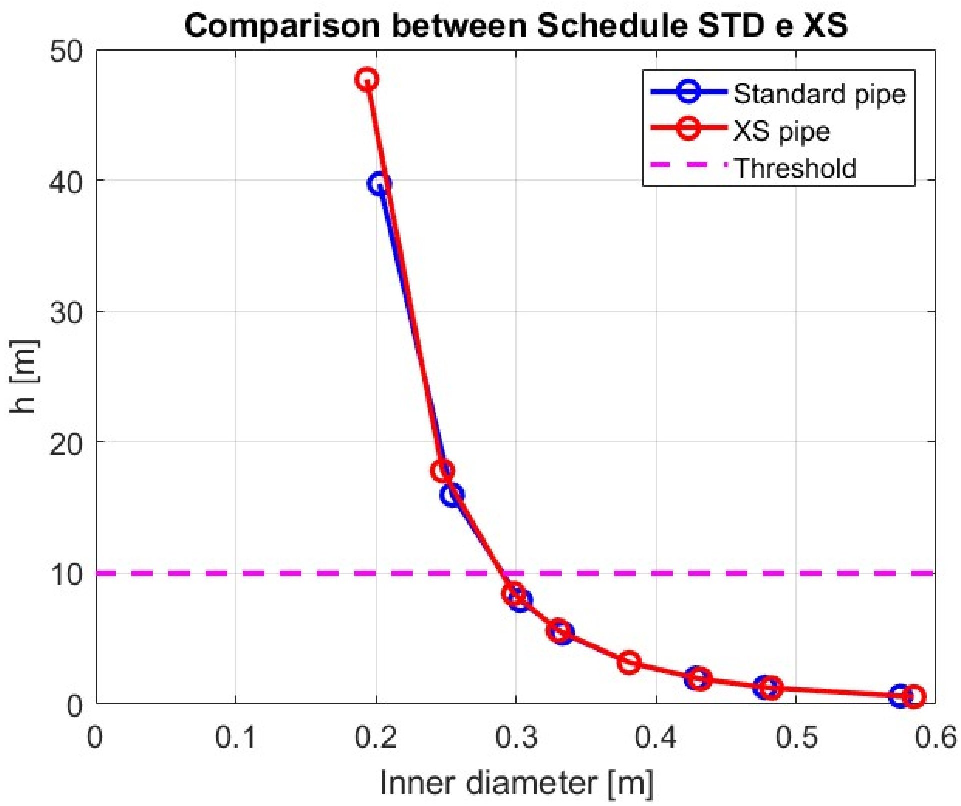

Figure 3: comparison between different diameters

Figure 3 illustrates the pronounced influence of pipe diameter on the required elevation head: increasing the diameter markedly reduces frictional losses, thereby enabling more compact layouts. Considering the 10 m design constraint, a pipe with an internal diameter equivalent to NPS 12 in (≈ 0.30 m, STD schedule) limits the required elevation to approximately 8 m, offering an optimal compromise between system compactness and hydraulic performance. Moreover, comparison of STD and XS schedules indicates that the thicker walled XS pipe by virtue of its slightly smaller internal diameter incurs marginally higher pressure drops; however, these differences remain modest under typical operating conditions.

2.3.2 Case with Variable Pipe Length: L=H

In this case, both pressure losses and buoyant driving force depend on the same geometric variable (the height, H), which leads to a non-linear relationship that must be solved iteratively.

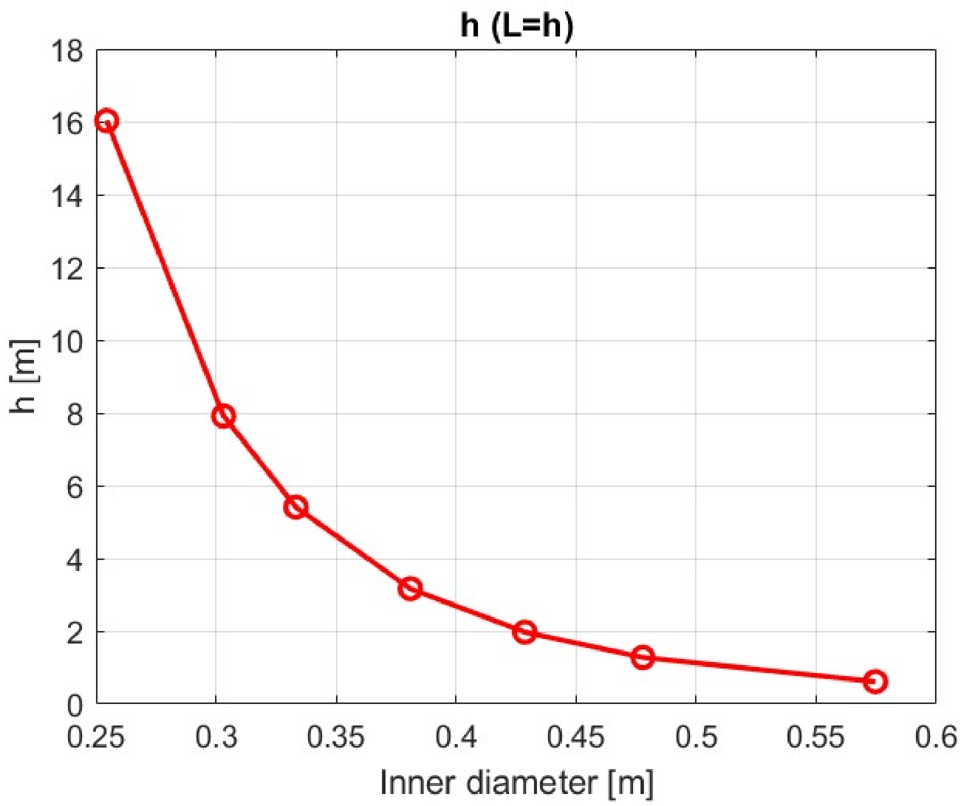

Figure 4: H=L

The result above confirm that the system remains well balances for NPS 12 in, with the corresponding value of H = L found to be approximately 7.9 m. This approach satisfies the thermal-hydraulic demands while ensuring a compact design, reducing vertical space requirements without compromising the performance of natural circulation.

2.3.3 Effect of Pipe Roughness on Natural Circulation

A 12-inch diameter pipe was selected, and the previous analysis was repeated for three scenarios: an ideal smooth-walled pipe and two industrially representative pipes with surface roughness values of $\epsilon = 2.5 \times 10^{-5}$ m and $\epsilon = 5.0 \times 10^{-5}$ m, respectively.

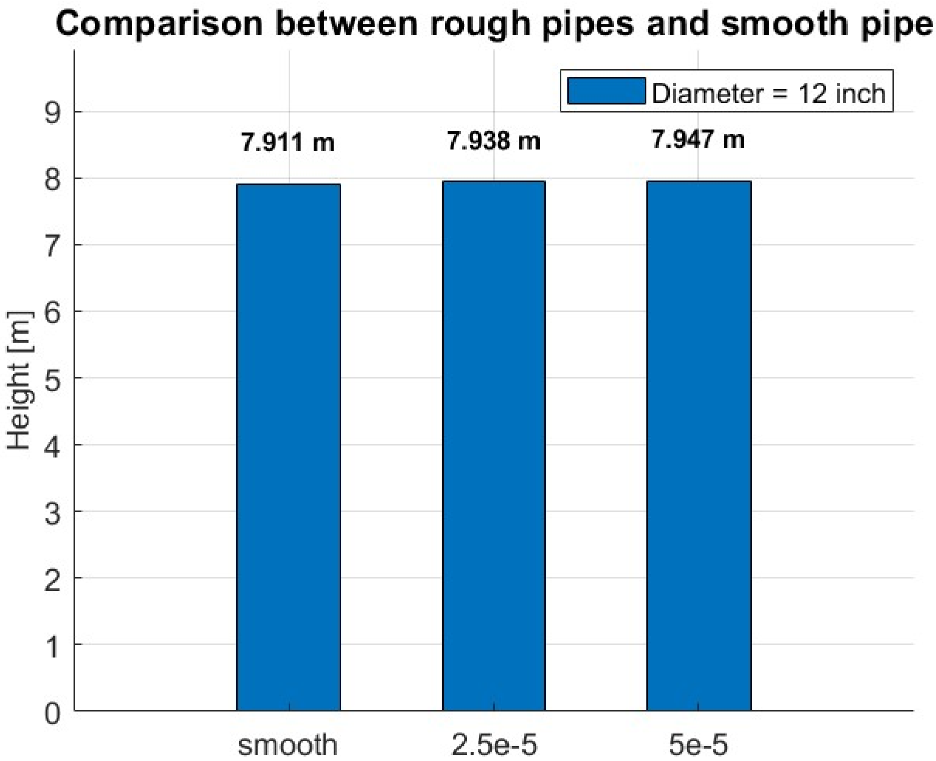

Figure 5: Height comparison between different roughness and smooth pipes

As shown in Figure 5, the additional elevation required to compensate for pipe surface roughness is negligible amounting to less than 1 % in the worst-case scenario. This outcome stems principally from the high Reynolds numbers in both the hot and cold legs, which render frictional losses relatively insensitive to variations in pipe roughness. Furthermore, since most of the total pressure drop is due to localized losses at the heater and cooler, the contribution of roughness to the overall hydraulic resistance is minimal.

In conclusion, the present analysis confirms that natural circulation offers a reliable heat removal mechanism for this configuration. The selected design utilizing NPS 12” piping and an overall elevation of approximately $8m$ maintains stable, effective performance across all evaluated conditions, validating its applicability for passive safety systems in nuclear reactors.

3 Assignment 1.2

The objective of this chapter is to analyze the decay heat removal system (DHRS), a passive safety system designed to remove residual heat from the reactor through natural circulation. The configuration under study consists of three interconnected circuits (Figure 6). The Primary System Circuit (PSC) extracts heat directly from the reactor core and transfers it to the first heat exchanger (HX1). HX1, in turn, exchanges heat with the Intermediate System Circuit (ISC), which subsequently transfers the thermal energy, through a second heat exchanger (HX2), to a cooling pool. This pool is connected to cooling towers through the Tertiary System Circuit (TSC).

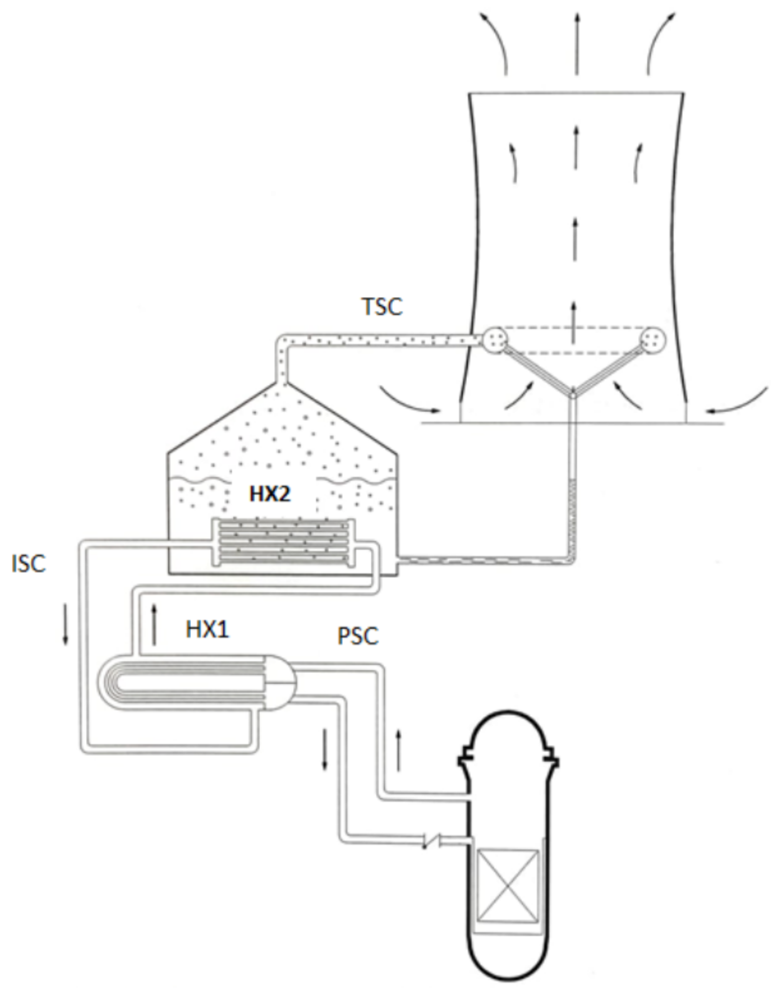

Figure 6: Passive Decay Heat Removal System of a fission reactors.

Once the system has been modeled, it becomes essential to determine the operating ranges of temperature and mass flow rate within the DHRS. This assessment requires a comprehensive thermal and hydraulic analysis of the system. In natural circulation systems, fluid motion is driven by density differences caused by temperature gradients at various points along the circuits. These density-driven flows are fundamental to the passive operation and effectiveness of the DHRS.

3.1 Assumption

Given the structure of the DHRS, the analysis begins from the **Intermediate System Circuit (ISC)** and progresses to the **Primary System Circuit (PSC)**. This order is chosen because the ISC is directly coupled to the passive cooling pool, which is maintained at known saturation conditions. Therefore, it provides more fixed parameters that allow for easier and more stable convergence.

The general assumptions used throughout this analysis are as follows:

- The system is in steady-state; transients after shutdown are neglected and the decay power removed is the 1% of the nominal thermal power;
- Constant pressure is assumed for each circuit: 70 bar for ISC, 75 bar for PSC;
- The cooling pool remains at 1 bar, saturation conditions;
- No fouling is considered in heat exchangers;
- HX1 is not a pure countercurrent heat exchanger: correction factor $F_T = 0.7$ is applied;
- To obtain accurate, temperature-dependent data, the thermophysical properties of water were determined using the XSteam library in MATLAB and the CoolProp library in Python.

3.2 Intermediate System Circuit

The analysis begins with HX2, as the pool-side boundary conditions are specified. In this heat exchanger, the ISC fluid flows through straight tubes immersed in a saturated pool. The pool-side heat flux is estimated using the McAdams correlation:

$$q'' = 2.257 \cdot \Delta T_{sat}^{3.86} \tag{12}$$

In this expression, q'' is the heat flux at the tube surface, and ΔT_{sat} is the temperature difference between the tube wall and the pool's saturation temperature.

$$h_{pool} = \frac{q''}{\Delta T_{sat}} \tag{13}$$

In this context, h_{pool} denotes the pool-side heat transfer coefficient of HX2.

The overall heat transfer coefficient U for HX2 is computed as:

$$U = \frac{1}{A_{out}\left(\frac{1}{h_{in}A_{in}} + \frac{\ln(D_{out}/D_{in})}{2\pi L k} + \frac{1}{h_{pool}A_{out}}\right)} \tag{14}$$

In this expression, h_{in} denotes the tube-side heat transfer coefficient of HX2; A_{in} and A_{out} are the internal and external heat transfer surface areas, respectively; D_{in} and D_{out} are the inner and outer diameters of the exchanger tubes; L corresponds to the tube length; and k is the thermal conductivity of the tube material.

Then, it has been evaluated the quantity of power that it has been transferred by the straight-tube heat exchanger, so it is used the log-mean temperature difference (LMTD) determined by:

$$\Delta T_{ml} = \frac{\dot{Q}_{th}}{UA} \tag{15}$$

Where $\dot{Q}_{th}$ is the steady state thermal power after the shut down and A is the external heat transfer surface area.

Subsequently it has been hypothesized the mass flow rate and by an iterative process the value of the hot and cold temperature has been evaluated. Below are the steps just listed for the calculation of T_{hot} and T_{cold}:

$$\dot{Q}_{th} = \dot{m}c_p(T_{h,i} - T_{c,i}) \tag{16}$$

In this relation, $\dot{m}$ is the mass flow rate, c_p denotes the fluid's specific heat at constant pressure, and $T_{h,i}$ and $T_{c,i}$ correspond to the hot-leg and cold-leg temperatures of the Intermediate System Circuit.

$$\dot{Q}_{th} = UA\Delta T_{ml} \tag{17}$$

$$\Delta T_{log} = \frac{(T_{h,i} - T_{sat}) - (T_{c,i} - T_{sat})}{\ln\left(\frac{T_{h,i}-T_{sat}}{T_{c,i}-T_{sat}}\right)} \tag{18}$$

Substituting equations (16) and (18) into (17)

$$\Rightarrow \frac{\frac{\dot{Q}_{th}}{\dot{m}c_p}}{\ln\left(1 + \frac{\dot{Q}_{th}}{\dot{m}c_p(T_{c,i}-T_{sat})}\right)} = \frac{\dot{Q}_{th}}{UA} \tag{19}$$

Assuming:

$$\alpha = \exp\left(\frac{\dot{Q}_{th}}{\dot{m}c_p\Delta T_{log}}\right) \tag{20}$$

$$\Rightarrow \frac{\dot{Q}_{th}}{\dot{m}c_p(T_{c,i} - T_{sat})} = \alpha - 1 \tag{21}$$

$$\Rightarrow T_{c,i} = T_{sat} + \frac{\dot{Q}_{th}}{\dot{m}c_p(\alpha - 1)} \tag{22}$$

The corresponding hot outlet temperature is:

$$T_{h,i} = T_{c,i} + \frac{\dot{Q}_{th}}{\dot{m}c_p} \tag{23}$$

3.2.1 Pressure Losses and Flow Calculation

Pressure losses include distributed and localized losses:

In fluid dynamics, distributed head losses originate from viscous shear stresses at the fluid–wall interface along the entire length of a conduit. These losses are categorized as major losses, since their cumulative contribution to the total head loss typically exceeds that of individual fittings or localized components.

Major losses:

$$\Delta P_d = f \frac{L}{D} \frac{\rho v^2}{2} \tag{24}$$

Where f is the friction factor.
Minor losses:

$$\Delta P_l = \sum_{i=1}^{n} k_i \cdot \frac{\rho_i v_i^2}{2} \tag{25}$$

In which k is the loss coefficient of each component.

Localized head losses, also known as minor losses, arise at individual circuit components (valves, bends, fittings, etc.). Each component's loss is quantified by its dimensionless loss coefficient k. Although conventionally classified as "minor," the distinction between "major" and "minor" losses does not always reflect their true impact: in systems with extensive piping runs with high density of components, the localized losses can exceed the distributed ones.

Shell-side k loss coefficient is computed from:

$$k_{shell} = 8 \cdot \frac{0.227}{Re^{0.193}} \cdot \frac{D_{shell}}{D_{eq}} \cdot (N_{baffles} + 1) \tag{26}$$

The total head loss is given by the sum of major (distributed) losses and minor (localized) losses:

$$\Delta P_{loss} = \Delta P_d + \Delta P_l \tag{27}$$

The driving force from buoyancy is:

$$\Delta P_{buoyancy} = (\rho_{c,i} - \rho_{h,i}) g H \tag{28}$$

In this context, $\rho_{h,i}$ and $\rho_{c,i}$ denote the fluid densities in the hot leg and cold leg of the Intermediate System Circuit, respectively.

By equating the net driving pressure head $\Delta P_{buoyancy}$ (28) to the total pressure drop ΔP_{loss} (27):

$$\Pi_i \, \dot{m}^2 = (\rho_{c,i} - \rho_{h,i}) \, g H_i \tag{29}$$

Where Π is equal to:

$$\Pi_i = \Sigma_i \left(f_i \frac{L_i}{D_i} \cdot \frac{1}{2\rho_i A_i^2} \right) + \Sigma_j \left(k_j \cdot \frac{1}{2\rho_j A_j^2} \right) \tag{30}$$

it is possible to obtain the mass flow rate as follow

$$\dot{m}_{new} = \sqrt{\frac{(\rho_{c,i} - \rho_{h,i}) g H_i}{\Pi_i}} \tag{31}$$

3.2.2 Results Summary: ISC

Table 1 summarizes the results obtained from the iterative code. In particular, the key parameters are the mass flow rate and the inlet and outlet temperatures of HX2. These values are essential for determining the mass flow rate and the cold-leg and hot-leg temperatures ($T_{c,i}$ and $T_{h,i}$) of the primary circuit, as required by the problem statement.

Parameter	**Value**
Hot Leg Temperature (HX2)	130.32 °C
Cold Leg Temperature (HX2)	108.67 °C
Mass Flow Rate (ISC)	65.53 kg/s
Distributed Loss (ISC)	386.01 Pa
Distributed Loss (HX2)	177.51 Pa
Bend Loss	631.93 Pa
Contraction Loss (HX2)	11.29 Pa
Expansion Loss (HX2)	22.09 Pa
Contraction Loss (HX1)	103.52 Pa
Expansion Loss (HX1)	178.16 Pa
Shell-side Loss	177.58 Pa
Total Localized Loss	1124.57 Pa
Total Distributed Loss	563.51 Pa
Total Pressure Drop (ISC)	1688.08 Pa

Table 1: Summary of key ISC values and head-loss contributions

3.3 Primary System Circuit (PSC)

Once the ISC analysis is complete, the output temperature of HX2 is used as input for HX1. To evaluate the heat exchanged in the U-tube heat exchanger it has been modeled using the corrected LMTD approach:

$$\Delta T_{ml} = \frac{\dot{Q}_{th}}{UAF_T} \tag{32}$$

Here, F_T denotes the correction factor introduced in the initial assumptions (section 3.1).

The outlet cold temperature is:

$$T_{c,p} = \left(\frac{\dot{Q}_{th}}{\dot{m}c_p} + T_{c,i} \cdot b - T_{h,i} \right) \frac{1}{b-1} \tag{33}$$

Where b is:

$$b = exp\left(\frac{U \cdot A \cdot F_T}{\dot{m}c_p} - \frac{U \cdot A \cdot F_T}{\dot{Q}_{th}} (T_{h,i} - T_{c,i}) \right) \tag{34}$$

By employing (16), one obtains:

$$T_{h,p} = T_{c,p} + \frac{\dot{Q}_{th}}{\dot{m}c_p} \tag{35}$$

3.3.1 Pressure Losses and Flow Calculation

The PSC is analyzed using the same equations and correlations developed for the ISC. Distributed and localized pressure losses are computed via those formula(24 and 25), and the buoyancy driven head is obtained from the density difference between hot and cold legs. An iterative procedure is then performed: initial temperature estimates, it set the thermophysical properties, from which the mass flow rate is calculated; updating flow rates it is computed the new temperatures, and iterations continue until both temperatures and flow rate converge within the prescribed tolerance.

For the Primary System Circuit, the calculation of the buoyancy-driven pressure difference also includes the reactor vessel, where pressure variations arise both from elevation differences and from the temperature gradient across the core. Accordingly, the core's contribution is evaluated as follows:

$$\Delta P_{core} = -g\left(\frac{\rho_h + \rho_c}{2}\right) H_2 \tag{36}$$

Total buoyancy:

$$\Delta P_{buoy} = \rho_c g(H_1 + H_2) - \rho_h g H_1 + \Delta P_{core} \tag{37}$$

Using the equation (29) the mass flow rate can be retrieved:

$$\dot{m}_{new} = \sqrt{\frac{\Delta P_{buoy}}{\Pi_p}} \tag{38}$$

3.3.2 Results Summary: PSC

Parameter	**Value**
Hot Leg Temperature (HX1)	149.22 °C
Cold Leg Temperature (HX1)	126.31 °C
Mass Flow Rate (PSC)	61.47 kg/s
Distributed Loss (PSC)	137.42 Pa
Distributed Loss (HX1)	740.08 Pa
Contraction (header–HX1)	27.39 Pa
Expansion (HX1–header)	53.60 Pa
Contraction (header–PSC)	91.99 Pa
Expansion (PSC–header)	167.01 Pa
Bend Loss	376.59 Pa
Valve Loss	24.85 Pa
Vessel Loss	44.28 Pa
Total Localized Loss	785.70 Pa
Total Distributed Loss	877.50 Pa
Total Pressure Drop (PSC)	1663.20 Pa

Table 2: Summary of key primary-circuit (PSC) parameters and head-loss contributions

Table 2 shows that frictional (distributed) losses are 877.5 Pa, account for approximately 52.8 % of the total head drop in the primary circuit, while localized losses contribute the remaining 47.2 % (785.7 Pa). Within the localized category, bend losses are the largest single contributor (376.6 Pa), whereas valve and vessel losses together represent under 5 % of the total.

4 Conclusion

The iterative model has demonstrated its effectiveness in establishing a consistent balance between the buoyancy induced driving head and the overall pressure losses, both distributed and localized. The resulting mass flow rates and temperature distributions are coherent with the passive safety performance criteria required for the DHRS. The analysis reveals that frictional losses represent the dominant resistance to flow, highlighting the importance to design properly parameters such as the diameter of the pipe and the roughness of the internal surface, to improve the system. On the other hand, the limited contribution of valve and vessel losses confirms the adequacy of the current component configuration. These outcomes support the capability of natural circulation to ensure reliable decay heat removal under the examined operating conditions.

NPS (in)	Outside Diameter (in)	Schedule												
		10	20	30	STD	40	60	XS	80	100	120	140	160	XXS
		Wall Thickness (in)												
1/8"	0,405				0,068	0,068		0,095	0,095					
1/4"	0,540				0,088	0,088		0,119	0,119					
3/8"	0,675				0,091	0,091		0,126	0,126					
1/2"	0,840				0,109	0,109		0,147	0,147				0,187	0,294
3/4"	1,050				0,113	0,113		0,154	0,154				0,219	0,308
1"	1,315				0,133	0,133		0,179	0,179				0,250	0,358
1 1/4"	1,660				0,140	0,140		0,191	0,191				0,250	0,382
1 1/2"	1,900				0,145	0,145		0,200	0,200				0,281	0,400
2"	2,375				0,154	0,154		0,218	0,218				0,344	0,436
2 1/2"	2,875				0,203	0,203		0,276	0,276				0,375	0,552
3"	3,500				0,216	0,216		0,300	0,300				0,438	0,600
3 1/2"	4,000				0,226	0,226		0,318	0,318					
4"	4,500				0,237	0,237		0,337	0,337		0,438		0,531	0,674
5"	5,563				0,258	0,258		0,375	0,375		0,500		0,625	0,750
6"	6,625				0,280	0,280		0,432	0,432		0,562		0,719	0,864
8"	8,625		0,250	0,277	0,322	0,322	0,406	0,500	0,500	0,594	0,719	0,812	0,906	0,875
10"	10,750		0,250	0,307	0,365	0,365	0,500	0,500	0,594	0,719	0,844	1,000	1,125	1,000
12"	12,750		0,250	0,330	0,375	0,406	0,562	0,500	0,688	0,844	1,000	1,125	1,312	1,000
14"	14,000	0,250	0,312	0,375	0,375	0,438	0,594	0,500	0,750	0,938	1,094	1,250	1,406	
16"	16,000	0,250	0,312	0,375	0,375	0,500	0,656	0,500	0,844	1,031	1,219	1,438	1,594	
18"	18,000	0,250	0,312	0,438	0,375	0,562	0,750	0,500	0,938	1,156	1,375	1,562	1,781	
20"	20,000	0,250	0,375	0,500	0,375	0,594	0,812	0,500	1,031	1,281	1,500	1,750	1,969	
22"	22,000	0,250	0,375	0,500	0,375		0,875	0,500	1,125	1,375	1,625	1,875	2,125	
24"	24,000	0,250	0,375	0,562	0,375	0,688	0,969	0,500	1,219	1,531	1,812	2,062	2,344	
30"	30,000	0,312	0,500	0,625	0,375			0,500						
32"	32,000	0,312	0,500	0,625	0,375	0,688								
34"	34,000	0,312	0,500	0,625	0,375	0,688								
36"	36,000	0,312	0,500	0,625	0,375	0,750								
42"	42,000		0,500	0,625	0,375	0,750								

- STD - Standard
- XS - Extra Strong
- XXS - Double Extra Strong
- NPS - Nominal Pipe Size

Figure 7: Pipe diameters from ANSI B36.10.